Applied Intelligence (2023) 53:3304–3317
https://doi.org/10.1007/s10489-022-03740-9

A process-centric performance management in a call center

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Accepted: 9 May 2022 / Published online: 27 May 2022
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Abstract
Discovering valuable information needs some extra focuses on business processes. Although data-centric techniques yield useful results, they are insufficient to explain the causes of the problems in the process. This study aims to reveal the relationship between customer satisfaction and other key performance indicators (KPIs) affected by the activities performed during the call process. The research applies process mining, a pragmatic analysis to obtain meaningful insights through event logs. Several statistical analyses also support the process mining to test the statistical significance. The study showed that customer satisfaction is positively affected by average handle time and first call resolution, whereas staff mistakes diminish it. Moreover, problem solving is much more important than waiting in the system. Waitlisted and Waitlisted back activities are crucial elements of a call center system. Moreover, the research presents an insight for customers who give the same score after the call. It explains not only KPIs' effects but also reasons for giving satisfaction scores based on call process. Additionally, in previous studies, the customer satisfaction indicator was mainly emphasized, but other KPIs' effects on satisfaction level were ignored. This paper evaluates the impact of the identified KPIs on satisfaction in a process-oriented manner.

Keywords Process mining · Customer satisfaction · Performance indicators · Call center · Customer relations

1 Introduction

The increase in the variety of product or service quality, the cheapening of the product or service with the same characteristics, and the strengthening of the competition in the sector have made necessary to gain new customers and increase the loyalty of existing customers. In order to win a customer, companies need to understand customers and meet their needs on time [24]. The sale of products or services to the newly acquired customer is five times more costly than selling them to the old customer [19]. On the other hand, customers want a timely, fast and solution-oriented response to their needs and wishes. For these reasons, companies try to understand customers by carrying out activities that add value to them and strengthening their loyalty with the company. One of these studies is the complaint management in call centers. Call centers, which are the bridge of communication between companies and customers, are the units that provide service by authorized staff at every hour of the day [9]. This situation creates a relationship that affects and nurtures each other between call centers and customer relationship management. They also contribute to the formation of customer loyalty by making the service more usable through the feedback they receive from the customer [8].

Customers' satisfaction with the provided service is the sum of their perception, evaluation, and psychological response during the experience. It is difficult to make the mental status and perception levels of the customers measurable by the company [7]. However, customers evaluate the service quality of the company by considering their interactions with the staff. The behaviors of call center employees and their ability to handle customer problems have a substantial impact on customer satisfaction [29]. When customers encounter a service that exceeds their expectations, they are satisfied even if no solution is available. However, if the customer's communication with the staff is negatively established, it leads to dissatisfaction with the customer. In this case, the company cannot evaluate customer satisfaction at a realistic and reliable level.

Various key performance indicators (KPIs) specific to call centers are considered in the literature and industrial

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case studies to evaluate process efficiency. Since the company where the case study implemented only aims to respond to customer needs, some KPIs based on sales, cost and other types of monetary metrics are excluded in the study.

Average handle time One of the most meaningful actions in KPIs for a call center effectiveness is to measure the average handle time. The handle time starts with the call center staff picks up the phone, and ends with disconnecting the call [38].

First call resolution It is another KPI to measure the effectiveness of a call center agent. First Call Resolution (FCR) refers to solve the customer problem during the first call. If a customer calls back more than one time or is transferred too often to another agent, it affects negatively system performance and customer satisfaction [37].

Customer satisfaction A common approach to test a call center efficiency is to measure customer satisfaction. Whereas other aforementioned KPIs can expose areas where the agents remain incapable, the customer satisfaction score is the most direct means to determine a call center performance to meet customer needs. Satisfaction is frequently checked by after-call surveys, notwithstanding other KPIs such as Average Time in Queue may be influenced during the call [7].

This study aims to reveal the relationship between customer satisfaction and other KPIs which are affected by the activities performed during the call. For this purpose, some statistical tests and process mining implemented by using call center event logs, and the flow within the system discovered. Firstly, the process flows of the surveyed, and non-surveyed customers were examined. Then, a new flow was created by taking into consideration the scores showing the satisfaction level of the customers participating in the survey. Under these research purposes, the study seeks answers to some primary research questions (RQs) by considering the predefined call center performance metrics.

- RQ1: Is there a statistical significance of call duration between surveyed and non-surveyed customers?
- RQ2: Is there a statistical difference between the flows of customers giving the full (5-point) and other scores?
- RQ3: How the flow and call duration affect the full (5 point) score of the surveyed customers?
- RQ4: Does agent workload impact call center system performance at peak time?
- RQ5: How satisfaction score is affected by other KPIs?

The remaining sections of the study are as follows: In Section 2, studies on call centers and process mining are presented. The differences between previous studies and this research are discussed. As the methodology used in the study, process mining is given in Section 3. Section 4 is the implementation part of this study. The data set used in the case study is introduced, and data pre-processing steps are given. RQs are answered one-by-one. Statistical tests are applied to see the differences between variables. Then, results of the process mining are discussed. Finally, findings and recommendations for process improvement is presented in Section 6.

2 Literature review

Many specific properties of service sector such as hard to standardization, stockless, and non-physical, make it difficult to analyze [26]. Since service quality, which is a multidimensional concept, directly affects customer satisfaction, companies should increase the quality of their services [11]. Customer satisfaction is the core to attain organizational success, especially in the service domain. Since it has growing interest to managers, it has been broadly investigated in recent years [6, 44, 49]. The basis of customer satisfaction is understanding what customers expect from the business. With the correct analysis of customer needs, companies increase customer satisfaction by providing the appropriate level of service.

A fast-growing number of customer satisfaction researches has applied data mining techniques such as sentiment analysis [27, 28, 36, 47] and latent semantic analysis [41, 50] as text mining approaches, classification [4, 5, 51], association rule mining [3, 6], and regression [32, 47] and proposed a new model to build high-level information for decision support [42]. Sentiment analysis is a popular data mining method used in customer satisfaction researches. It is a kind of text mining approach considering the reviews and emotions involved in the text, which mainly contains online customer reviews about products or services. Wang et al. [47] tried to understand from the customer opinion with sentiment analysis if a customer is satisfied or not. Then, they estimated how product properties affect customer satisfaction levels using logistic regression. Kang and Park [28] collected and pre-processed real voices of customers who state comparatively unambiguous reviews. They defined sentiment words and measured customer satisfaction scores with a multi-criteria decision-making technique. Jung and Suh [27] focused on measuring job satisfaction determinants of agents through online agent opinions. Aktepe et al. [4] evaluated the satisfaction and loyalty level of customers under changing market conditions with classification and structural equation modeling by interviewing face-to-face. Table 1 presents a summary for comparison of previous studies.

Table 1 Comparison of previous studies

Study	Analysis	KPI	Perspective	
			Customer	Agent
[28]	Sentiment analysis	Satisfaction	YES	NO
[47]	Sentiment analysis	Satisfaction	YES	NO
[27]	Text mining	Satisfaction	NO	YES
[4]	Classification and SEM*	Satisfaction	YES	NO
[7]	SEM*	Satisfaction	YES	NO
[33]	Descriptive statistics	Satisfaction and more	YES	NO
[20]	Descriptive statistics	Satisfaction and more	YES	NO
[25]	Descriptive statistics	Satisfaction and more	YES	NO
[48]	Process Mining	N/A	NO	NO
[30]	Process Mining	N/A	NO	NO
[34]	Process Mining	Satisfaction	YES	NO
This study	Process Mining	Satisfaction and more	YES	YES

*SEM: Structural Equation Modeling

Customer satisfaction gains more importance at call centers. Call centers where satisfaction score is mainly measured as one of the most critical KPIs are departments that aim to understand and meet customer requirements, wishes and problems. They provide vital services for customer satisfaction, which is a complicated area [39]. Chicu et al. [7] aimed to explore the human-based factors that affect the satisfaction level in call centers using structural equation models. In fact, the caller satisfaction level is affected by not only human-based factors but also factors out of control of call center activities such as quality, organization strategy, target markets, customer mood, decision-making processes [2]. However, some researches concentrated on the interaction between first call resolution and satisfaction level because first call resolution positively mediates the relationship between caller satisfaction and knowledge management. Manzoor and Shahabudeen [33] expressed that customer satisfaction score is affected by FCR positively, whereas it affected negatively by average handling time. Feinberg et al. [20] investigated the operational elements of customer satisfaction in the call center. They evaluated the relationship between customer satisfaction and critical variables experimentally. Jaiswal [25] studied customer satisfaction with the tools used to measure service quality in call centers.

As an up-to-date technique that can be used for performance management of call centers, process mining techniques can yield meaningful solutions by enabling real insights related to the actual processes due to its fundamental principle on the executed and stored event logs. Many studies focused on service industry by using process mining method for various purposes such as understanding customer behaviors [15, 17, 21], investigating process-oriented health problems [13, 35, 40], uncovering the interaction between people [1], and customer segmentation from their paths [18]. Since service industries are mainly human-centric, business processes are affected by behaviors of staffs [12, 14]. Wongvigran and Premchaiswadi [48] followed the interactions between different crews in a functional organizational structure from top to bottom by process mining. The proposed approach presents to reduce customer dissatisfaction by detecting possible inconsistencies arising from the previously defined rules and responsibilities. Lamine et al. [30] used process mining to obtain an overview of emergency processes. They used discrete event simulation for different what-if scenarios to address the potential problems of the emergency call center. Panpanich et al. [34] applied social networking process mining techniques to call center data to examine the behavior of customers who contacted the call center department. They also identified the bottlenecks in the process and possible staff errors.

When compared with other approaches in the call center field, process mining has some advantages. Since process mining applies data mining techniques for process management tasks, it benefits from the advantages of data-based methods. Moreover, it can give an idea about the whole system instead of a part of it. It considers the relationships between activities in the process while producing valuable solutions. Objectivity, speed, little effort, and completeness are other benefits of process mining [10]. However, it has also some drawbacks. One of the most encountered shortcomings is data access and preparation. Finding and preparing the right data (event

log) for the process analysis. Techniques and tools require to be more integrated with other analysis methods [46]. Under these comparisons, the main focus of the study is to measure the level of customer satisfaction under process-based research rather than only data-driven results. Previous studies focus on the satisfaction scores but they cannot explain the reasons. This study presents an insight for customers who give the same score after the call. Some suggestions are provided to improve system performance under process mining results. Additionally, in previous studies, only the customer satisfaction indicator was emphasized, but the effects of other KPIs on satisfaction level were ignored. This paper evaluates the impact of the identified KPIs on satisfaction in a process-oriented manner.

3 Process mining

Process Mining feeds on information systems that provide business intelligence to convert information into actionable knowledge [31]. Process mining based on intelligent systems refers mainly to implementing intelligent systems-based approaches to find the current patterns in the information [23]. Process mining technology uses intelligent systems to automatically extract, visualize, and make sense of operational data from organizations' IT systems. It uncovers root causes and can predict future problems to take action proactively. So, managers can make data-based, quick, more reliable, and comprehensive decisions. With this perspective, process mining and intelligent systems work together to improve business processes through intelligent techniques.

Discovery algorithms in process mining create graph-based process models [22]. Call log analysis is one of the application fields of graph-based process models that proposes to perform advanced information straightforwardly and understandably. They can be regarded as a sort of Bayesian network, which holds the traceability of all the real event logs in the nodes and transitions. Hence, researchers have a chance to retrieve event sets using these nodes and transitions.

The goal of using an event log to find process timing provides a well-defined, reliable, and unobtrusive way to gauge the performance of process models. The duration of significant process steps and detailed developments can be analyzed using IT systems [43]. However, simple statistical methods over the business processes might not be acceptable to perform an overall human understanding of the process performance. Recent techniques are required to benefit from the event log to present a comprehensive picture of the performed process. This is the case of why process mining is chosen and applied in this study.

Process mining (PM) is a method that aims to identify the problems in real processes rather than the constructed processes with event logs by establishing the link between traditional process science and data science methods [15]. The main objective of process mining is to discover, monitor, and improve the actual processes. Each event log, which is inputs of process mining, contains information about the activities in the process such as time, person, cost. Enterprises can quickly obtain event logs from IT systems such as ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), SCM (Supply Chain Management) [45]. Table 2 presents a sample of event log. Three primary features should be in the event log: Case ID, Timestamp, Activity. Some additional information such as the resource performing the activity and the cost of this source can be added to the event log [16]. The extension number is included in this survey as an additional attribute since it represents the staff who answered the call.

PM establishes log-based ordering relations among activities [45]. It is said that there is a causal dependency between activities a and b if activity a is followed by activity b, and activity b is never followed by activity a.

Let L be an event log on A and let $a, b \in A$. $a >_L b$ if and only if there is a trace $\sigma = \langle t_1, t_2, t_3, ..., t_n \rangle$ and $i \in 1, ..., n-1$ such that $\sigma \in L$, $t_i = a$ and $t_{i+1} = b$; $a \rightarrow_L b$; if and only if $a >_L b$ and $b \not>_L a$; $a \#_L b$; if and only if

Table 2 A sample of event log

Case ID	Event ID	Timestamp	Activity	Attributes		
				Resource	Cost	...
1	423512	05-10-2019:10.05.38	Ringing	Dogan	150	...
	423513	05-10-2019:10.06.49	Taking the call	Deniz	200	...
	424214	05-10-2019:10.06.50	Answering	Defne	50	...
	426515	05-10-2019:10.09.36	Ringing off by staff	Onur	200	...
2	426521	05-10-2017:11.32.32	Withdrawn	Defne	50	...
	426522	05-10-2017:11.32.33	Answering	Defne	50	...
	426524	05-10-2017:11.35.01	Ringing off by customer	Onur	200	...

$a \not>_L b$ and $b \not>_L a$; $a||_L b$; if and only if $a >_L b$ and $b >_L a$. For instance, $L = [\langle k, l, m, n\rangle^3, \langle k, m, l, n\rangle^2, \langle k, o, n\rangle]$ gives the traces in the log L. The power of the traces shows the number of counted traces in the same order. In other words, $\langle k, l, m, n\rangle$ is confronted three times in the log L. For this event log, the log-based ordering relations can be created as follows:

$$>_L = \{(k, l), (k, m), (k, o), (l, m), (l, n), (m, l), (m, n), (o, n)\}$$

$$\rightarrow_L = \{(k, l), (k, m), (k, o), (l, n), (m, n), (o, n)\}$$

$$\#_L = \{(k, k), (k, n), (l, l), (l, o), (m, m), (m, o), (n, k), (n, n), (o, l), (o, m), (o, o)\}$$

$$||_L = \{(l, m), (m, l)\}$$

A footprint matrix like given in Table 3 is created to show relations among activity pairs. The relation $>_L$ indicates all activity pairs in a *direct follow*. For instance, k is directly followed by l in the trace of $\langle k, l, m, n\rangle$ and shown by $k >_L l$. Besides, $l \not>_L k$ since k does not directly follow l in any trace. The relation $\rightarrow_L$ presents whole relation activity pairs with a *causality relation*. $m \rightarrow_L n$ since n sometimes directly follows m and m never directly follows by n. The relation $\#_L$ gives all activity pairs that do not directly follow each other. Because l is never followed by o and never the other way around, given by $l\#_L o$. The relation $||_L$ refers parallel activity pairs. $l||_L m$ since sometimes l directly follows m and sometimes vice versa. In other words, it is presented by $l >_L m$ and $m >_L l$.

4 Case study

In this study, call center processes of a company in the energy sector was examined. Data were taken from the company's Subscriber Information Management System (ABYS). 42185 call events were used for process mining. Firstly, data were obtained from ABYS and some data preparation steps were conducted. Under defined RQs, some statistical tests were applied. Then, process mining was implemented. The visualization of the discovered process maps of the customers participating and not participating in the satisfaction survey was given. Additionally, the process flows of customers given 5 points and other points were presented to discuss aspects that affect customers' evaluation on the service quality. The study considers peak time call processes and agent workload. Additionally, predefined KPIs were calculated to show the affect on satisfaction score.

Table 3 Footprint of event log L

	k	l	m	n	o
k	$\#_L$	$\rightarrow_L$	$\rightarrow_L$	$\#_L$	$\rightarrow_L$
l	$\leftarrow_L$	$\#_L$	$||_L$	$\rightarrow_L$	$\#_L$
m	$\leftarrow_L$	$||_L$	$\#_L$	$\rightarrow_L$	$\#_L$
n	$\#_L$	$\leftarrow_L$	$\leftarrow_L$	$\#_L$	$\leftarrow_L$
o	$\leftarrow_L$	$\#_L$	$\#_L$	$\rightarrow_L$	$\#_L$

4.1 Data pre-processing and basic statistics

Since the obtained data were not suitable for process mining, some data pre-processing steps were performed. Data fields that will not be used in the research were removed. Many IT systems have single time information (only start or end time) related to activities. In this study, the end time was obtained by adding the call duration to the start time. In another preparation step, repeated event logs were excluded from the dataset with the same Call ID, taking into account the start time from the first event and the end time from the last event. There are six data attributes in the prepared event log: Call ID, Start, End, Activity, Extension (Ext.), and customer satisfaction score (CSS). The Start and End columns indicate the start and end time of the call event logs. The Activity column describes the operations performed during the call. The Extension column shows the extension telephone number of the staff answering the call. It also represents the staff. The customer satisfaction survey is answered in the range of 1 - 5. A zero-valued CSS indicates that the customer connected to the call center did not participate in the satisfaction survey.

During the research date range, 333 customers contacted the company using the call center and answered survey questions. Satisfaction survey records are stored in a different system. However, there are no call details in the satisfaction survey. Only the results of the survey are stored. The two databases were integrated by establishing a relationship between the end time of the call and the start time of the survey like shown in Fig. 1.

4.2 Response to research questions

4.2.1 RQ 1: Is there a statistical significance of call duration between surveyed and non-surveyed customers?

This RQ includes all customer calls. The purpose of the question is to learn whether there is a significant difference between customers participating and declining to survey in terms of call duration.

H_0: The average duration is equal for surveyed and non-surveyed customers

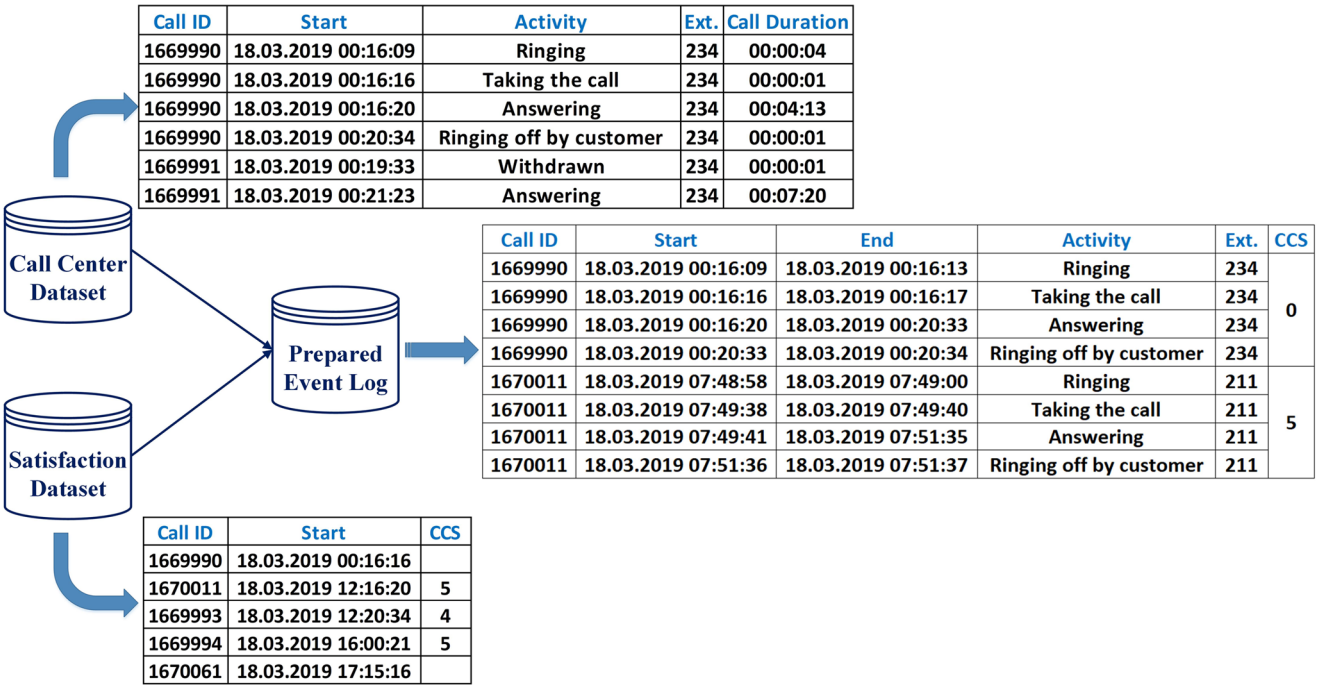


Fig. 1 Integration of data from different systems for process mining

H_1: The average duration is not equal for surveyed and non-surveyed customers

Because the normal distribution is an assumption for the test of statistical differences between numerical and categorical variables, call duration as a numerical variable and participating satisfaction survey as a categorical variable were tested for normality. Figure 2 gives statistical results. The Kolmogorov-Smirnov test indicates that call duration does not normally distribute because both significance levels are less than the alpha error rate, which is 0.05. As a non-parametric test, the Mann-Whitney U test was then applied to compare whether there is a statistically significant difference between call duration considering participating satisfaction survey or not. A less significance level than 0.05 refers to the significant difference. This means the null hypothesis is rejected and the average call duration is different for surveyed and non-surveyed customers.

4.2.2 RQ 2: Is there a statistically difference between the flows of customers giving the full (5-point) and other scores?

This research question considers only surveyed customers calls and divides these customers into two groups as giving full score and other scores as satisfaction point. The purpose is to explain the status of customers' satisfaction level through activities in the call process.

H_0: The average duration is equal for customers giving full score and other scores

H_1: The average duration is not equal for customers giving full score and other scores

In Fig. 3, The Kolmogorov-Smirnov test indicates the average duration has not a normal distribution. Therefore,

Fig. 2 Statistical tests for surveyed and non-surveyed customer

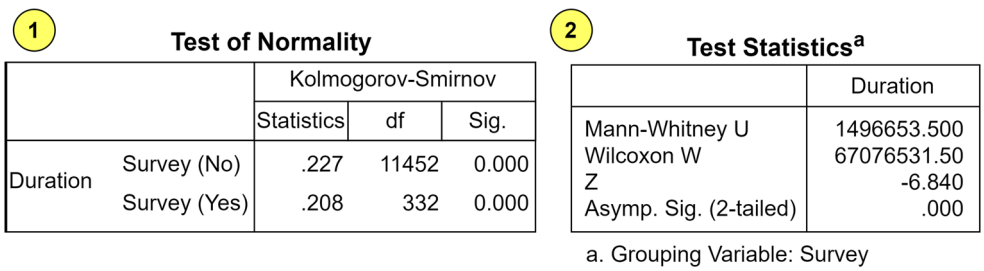


Fig. 3 Statistical tests for customer with full and other score

1

Test of Normality

		Kolmogorov-Smirnov		
		Statistics	df	Sig.
Duration	Score (Other)	.258	37	0.000
	Score (Full)	.200	295	0.000

2

Test Statistics[a]

	Duration
Mann-Whitney U	4677.000
Wilcoxon W	5380.000
Z	-1.456
Asymp. Sig. (2-tailed)	.145

a. Grouping Variable: Score

the Mann-Whitney U test was applied by grouping score input. The null hypothesis, H_0 is not rejected because the asymptotic significance value 0.145 is greater than alpha value 0.05. This means the difference between customers giving full score and other scores. In other words, satisfaction scores do not depend on call duration for customers participate in the survey.

4.2.3 RQ 3: How the flow and the call duration affect the 5 points (full score) of the surveyed customers?

Since RQ 2 indicates the statistically non-significant difference between customers giving full score or other scores, Figs. 4 and 5 was created by fuzzy miner algorithm, a process mining discovery algorithm, to visualize the similarities of the flow and call duration on satisfaction score. The thickness of the arrows increases when the number of transitions between two nodes increases. Similarly, the darkness of the colors in the nodes increase when the number of executed activities increases.

Figure 4 shows that 7195 unique customers who gives full points to the survey started with the "Ringing" activity, whereas only five processes started with "Operator did not answer". If the agent does not see the call or cannot press the answer key, the call may not be answered. In such cases, the agent calls the customer back as soon as possible. The so-called "Operator did not answer" activity occurs when the computer is busy, or there is a systemic problem.

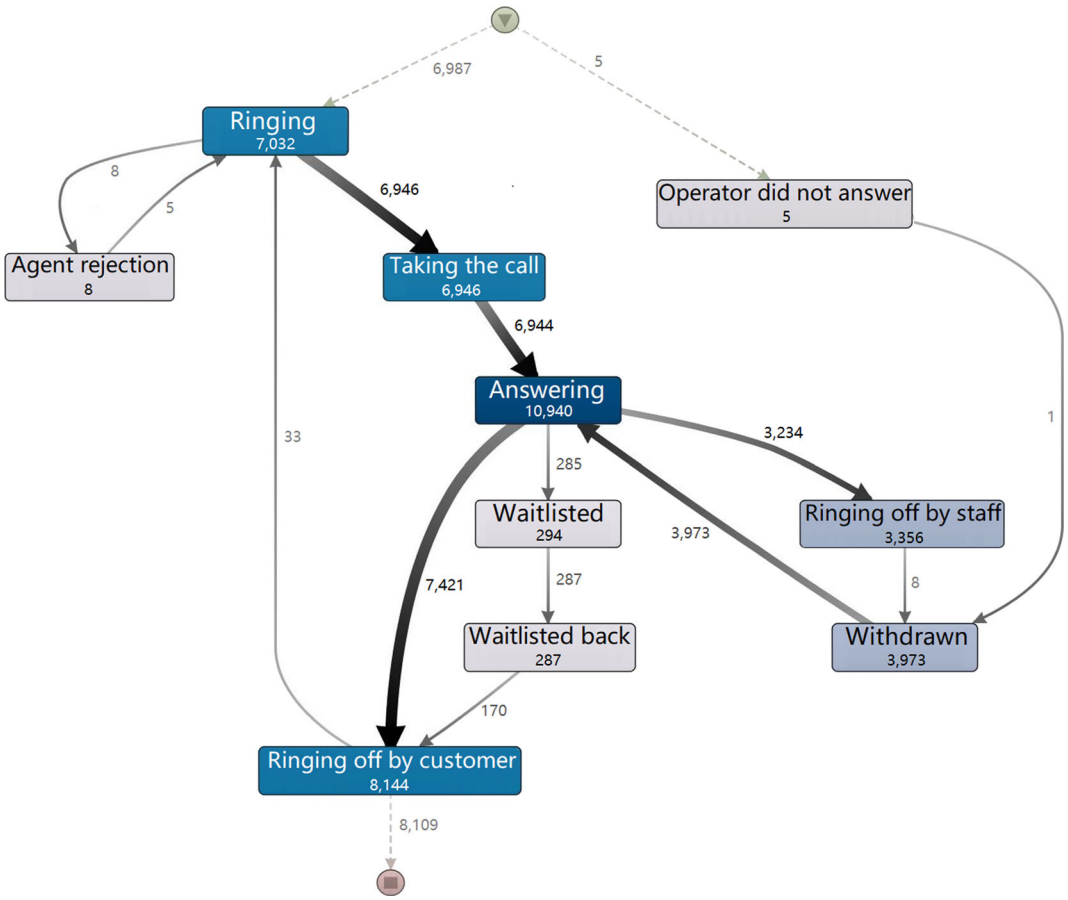


Fig. 4 Discovered process flows for 5-point customers

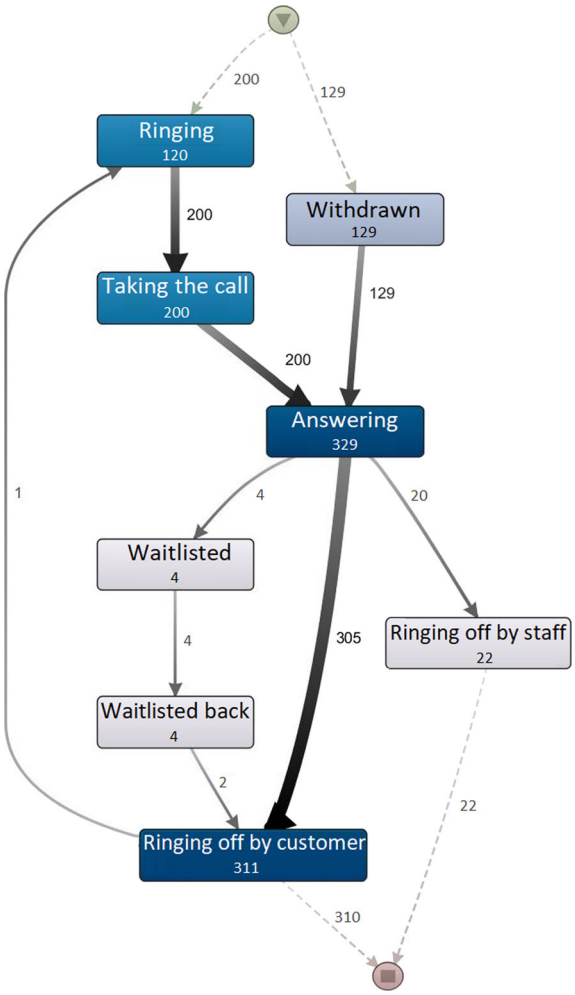


Fig. 5 Discovered process flows for other-point customers

Customers who cannot connect any personnel are withdrawn later. After "Ringing" activity, one of the agents takes the call and answers the calling. There are three options after answering the call: i) ringing off by staff, which is expected to continue with "Withdrawn" activity, ii) customer waits for a while ("Waitlisted") because the agent needs more information about the question, iii) Ringing off by customer, which is the expected end. The company has some scenarios for these critical activities shown in Fig. 4. The "Ringing off by staff" activity is based on the scenario where the staff asks if there is an issue that can help the customer before ending the call, and if no response is received, the staff can terminate the call. As the second scenario, "Withdrawn" means that agent can accept the customer by answering the call without waiting for after call work time, nine seconds. Another critical activity in the flow is "Waitlisted" activity. If the staff does not have enough information to meet the customer's needs, they consult with a experienced person. In this case, the call is put on hold and returns to the call after a while. It is reflected in the system as "Waitlisted back". "Ringing" activity is directly occurred after "Ringing off by customer" activity in 33 out of 8144 cases by calling customer agent again.

Figure 5 shows the flow of other-point customers. As in Fig. 4, the trace of $>_L = \langle$ Ringing, Taking the call, Answering, Ringing off by customer $\rangle$ is the most repetitive path. This flow also shows the dominant trace followed by customers who answered the satisfaction survey. However, 200 case studies started with the "Ringing" activity, while 129 of them started with "Withdrawn", which increases the average handle time, one of the KPIs considered in this study. The call process ends with either "Ringing off by the customer", which is frequently occurred or "Ringing off by staff".

Figure 6 presents the process models of the satisfaction scores of customers. Figure 6a shows the flow of customers giving full point to the provided service; Fig. 6b gives the flow of customers giving other scores (from 1 to 4). The followed process steps in Fig. 6 are the same in Figs. 4 and 5. Some activity times (such as "Ringing" and "Taking the call") and processing times between activities (such as from "Waitlisted" to "Waitlisted back") neglected due to ignorable (or instant) duration. Therefore, in some cases, the darkness of the activities (nodes in rectangles) and the thickness of the arrows do not give a meaningful result. The average call duration is 3.2 minutes for Fig. 6a and 3.8 minutes for Fig. 6b. The "Answering" activity has the largest average expectedly for both process models. It is seen that the customers who give 5 points wait for an average of 49 seconds in the system.

Table 4 shows the statistics of the full scored customers. It is seen that the activities with the highest average duration are "Answering" and "Waitlisted back" for customers with full-scored and giving other scores, respectively.

4.2.4 RQ4: Does agent workload has an impact call center system performance at peak time?

The peak time is a period that should be investigated because of overcrowding. Figure 7 presents time-based changes in the call center. Blue lines show the normal period in the call center, whereas orange lines present the peak time. One out of every five calls started at 11.00 and 13.00, which indicates the peak time. Almost no call received before 07.00. There is a dramatically decrease after 17.00 and downwards continues at the remaining hours of the days. There are two peak times, one is before noon, and the other is afternoon. The midday corresponding to the lunchtime is the local minimum point.

This RQ includes both surveyed and non-surveyed customer calls. First of all, the statistical significance was tested for the peak time to investigate whether agents affect the system performance under KPIs defined before. Hence,

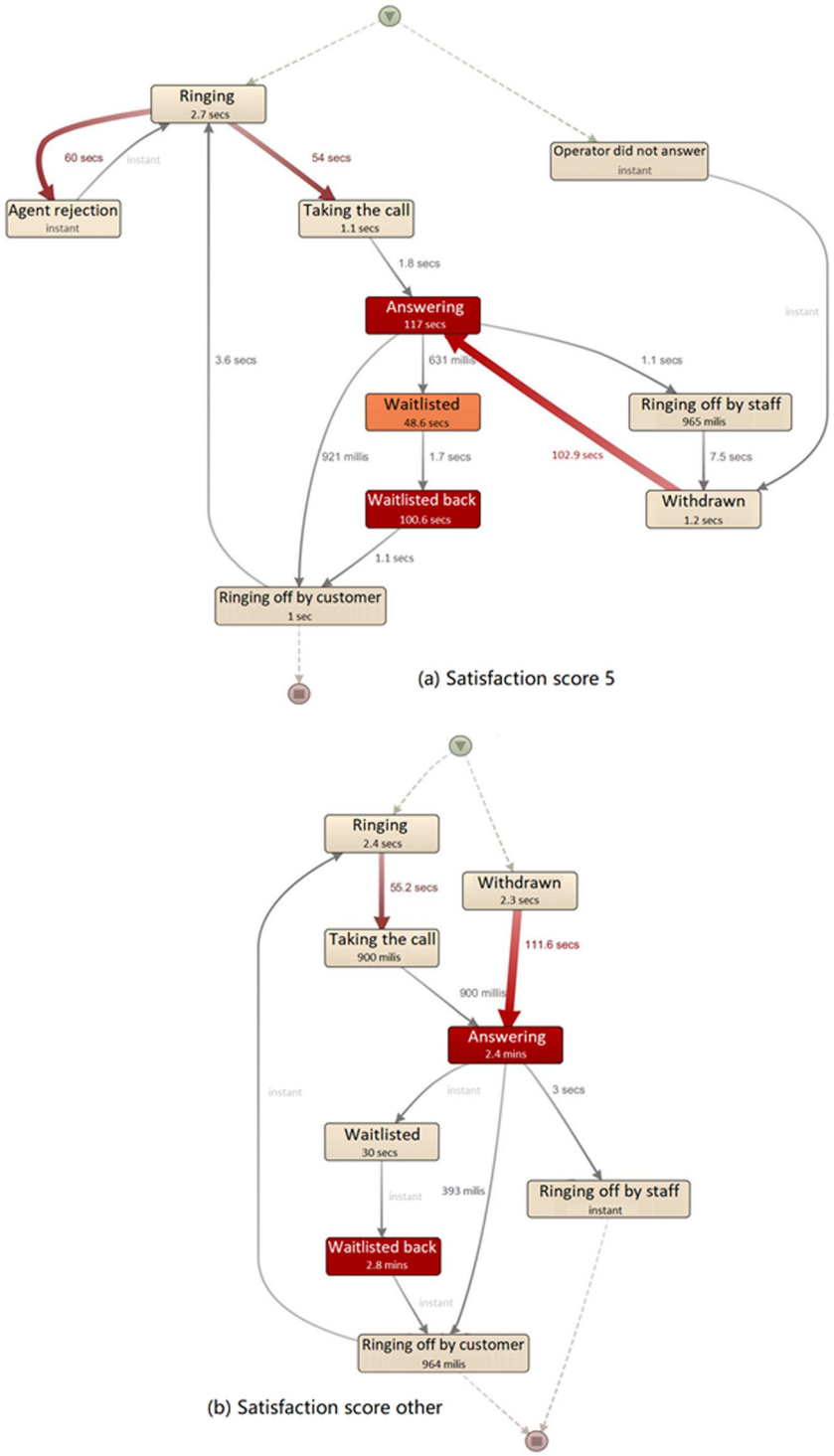


Fig. 6 Average time-focused process flow of surveyed customers

the following null and alternative hypotheses were applied to test firstly the differences of normal and peak time.

H_0: Average call duration is similar during normal time and peak time

H_1: Average call duration is not similar during normal time and peak time

Figure 8 concentrates on deviations from the normal (reference) distribution. The straight line in the plots

Table 4 Time-based critical process metrics for surveyed customers

Process metrics and critical activity	Full score	Other score
Highest average duration	117 seconds	168 seconds
Activity with the highest duration	Answering	Waitlisted back
Answering activity	117 seconds	144 seconds
Waitlisted	48.6 seconds	30 seconds
Waitlisted back	100.6 seconds	168 seconds

presents the perfectly normal distribution. It is clear that both normal time and peak time has no normal distribution. For two independent samples test, the Mann-Whitney U test was applied to see the equality of two distributions. The asymptotic significance value (< 0.001) was less than alpha value 0.05. Therefore, the null hypothesis should be rejected. This means average duration at peak time differs from normal time. Table 5 demonstrates the workload density for staff during peak time because agent workload at peak time affects the call center system performance. Two out of 13 agents with extension number 225 and 235 mainly answer the incoming calls in the call center. 48% of customers contact with these two staffs and the average duration staying on the hold between 2.5 minutes and 3 minutes.

Since the performance of staff 225 and staff 235 is critical, their processes are investigated in Fig. 9. These agents start to a call with either withdrawing a customer or "Ringing" activity. Then, they answer the customer needs. Finally, the majority of calls end with "Ringing off by the customer". A call rarely finishes with "Ringing off by the staff" and directly ends after "Taking the call" before "Answering". The two overloaded staffs have an expected flow during the calls. They either answer regular calls or withdraw customers waiting in the system. Interestingly, there is no "Waitlisted" activity, that is an indicator of the well-qualified staff. As a normal flow, only one call is terminated by the staff. Almost all customers end the call after their needs are met.

4.2.5 RQ5: How satisfaction score is affected by other KPIs?

The key performance indicators are computed and given in Table 6. Suitable cases (totally 7195 unique customers) include non-transferred customer calls. Therefore, FCR is calculated by ignoring multiple customer calls for the same customers. FCR affects caller satisfaction positively. Customers who give full score have a higher FCR rate with 0.60, whereas other-point customers have 0.22 FCR rate. These results support previous studies [33]. However, average handle time does not affect negatively in this case study. Although Manzoor et al. [33] declared it has negative impact on satisfaction score, it is not valid for all case studies such as ours. Some customers may feel that the company tries more to solve the problems and conclude that they have cared more and valued. Hence, it may have given 5 points in the satisfaction survey. In other words, the customer needs to solve the problem with the company, which is more significant than some extra minutes.

The difference between median and mean handle time shows deviation of call performance. Whereas 5-point customer processes have a 0.1 points difference, other-point customer processes have a higher difference with 0.8 points. This means processes of customers giving less than 5 points include some outlier data that result in difference between average handle times.

5 Managerial implications

Some insightful implications are presented as follows to summarize the study results. RQ 1 showed that the average duration is not distributed normally and differs for surveyed and non-surveyed customers. Because of the importance of customers' experience in contact with agents, managers should encourage customers by considering call duration to increase participation in the questionnaire.

If we focus on the customers surveyed, although call duration affects participation in the questionnaire, according to the RQ2, it is not important statistically for satisfaction

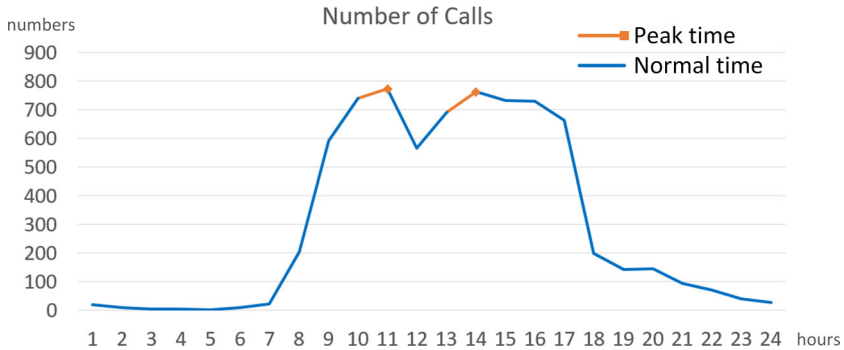


Fig. 7 Time-based changes on average

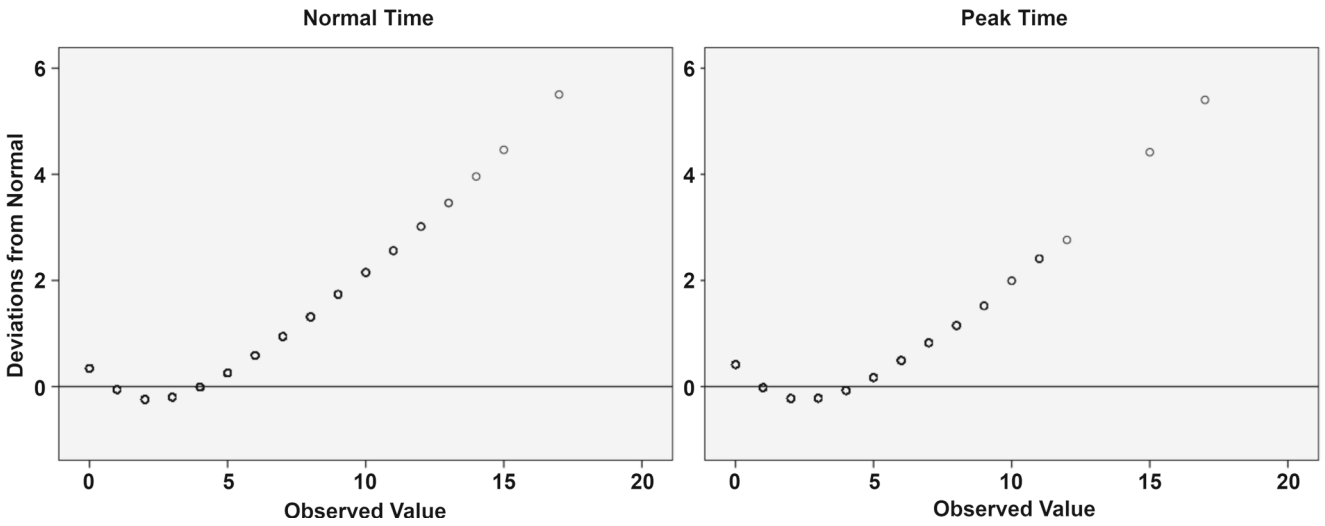


Fig. 8 Normality test of normal time and peak time

scores. Hence, managers should consider call duration to raise the participation rate but look at the call process to boost satisfaction.

This study analyzes the customers' call process to determine how to increase satisfaction scores with RQ3. Even though the most dominant trace is the same (⟨ Ringing, Taking the call, Answering, Ringing off by customer ⟩) for both customers giving a full score or other scores, the 'Withdrawn' activity is essential to get a full score. The flow of customers scoring less than 5 indicates that 40% (129 out of 329) of the calls start by withdrawing a customer. Therefore, to get a full score, agents should answer calls when they receive them without becoming unanswered, resulting in a withdrawal activity.

The crowdedness of incoming calls may affect the agents' performance, resulting in fewer satisfaction scores. To make this point explicit, this study considers the agent's workload impact on satisfaction scores with RQ4. It is presented that the average duration at peak times differs is not similar to those of regular times. Managers should balance the workload of staff to get a higher satisfaction score. In our case, two agents answer about one of every two calls, and their call process usually ends up with less score.

Although satisfaction score is the most highlighted KPI, the relationship between it and other KPIs related to a call center process is considered in this study with RQ5. i) There is a direct proportion between satisfaction score and first call resolution. Managers should train staff periodically to be able to avoid not solving the customer's problem on the first call. ii) It is expected that the average handle time negatively affects the satisfaction score [33]. However, this study indicates that there is no standing relationship between the average handle time and satisfaction score.

Table 5 Workload density statistics

Resource (External)	Frequency	Relative Frequency	Mean Duration	Duration Range
225	21	0.3	2.9	21
235	13	0,18	2.54	17
241	7	0,10	3	12
223	6	0,08	2.83	7
242	5	0,07	2.8	4
202	4	0,06	1.5	6
243	3	0,04	5.33	6
234	3	0,04	3.67	11
211	2	0,03	4	8
210	2	0,03	3	6
201	2	0,03	1.5	3
209	2	0,03	1	2
207	1	0,01	3	3

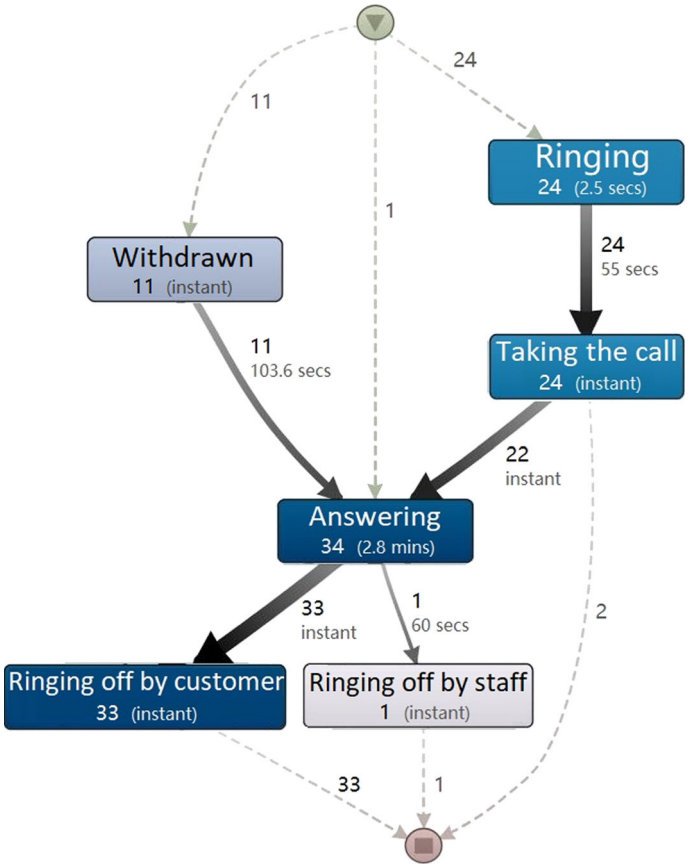


Fig. 9 Staff workload during peak time processes

In a call center, personnel mistakes decrease customer satisfaction. "Waitlisted" and "Waitlisted back" activities are crucial elements of a call center system. On contrary to the expected situation, waitlisted customers give the highest points in the energy domain. Problem solving is much more important than waiting in the system. These customers may feel that the company tries more to solve the problems. Many previous types of research showed handle time has a negative relationship with satisfaction. However, average handle time (and also first call resolution), which are other performance indicators for a call center process, positively affects customer satisfaction in the energy domain.

Table 6 Key performance Indicators (KPIs)

	Other points	5 points
Suitable calls	268	6927
Median Handle Time (min)	2.6	3
Mean Handle Time (min)	3.2	3.1
Total Calls	1200	11452
Median Handle Time (min)	3.0	3
Mean Handle Time (min)	3.7	3.2
First Call Resolution	0.22	0.60

6 Conclusion and improvement suggestions

This study aims to create process-oriented evaluations for customer satisfaction in a call center. For this purpose, call center event logs stored in the IT systems were used as input for process mining. Process mining was used for five research topics. i) Is there a statistical significance of call duration between surveyed and non-surveyed customers?, ii) Is there a statistical difference between the flows of customers giving the full (5-point) and other scores? iii) How the flow and the call duration affect the full score of the surveyed customers?, iv) Does agent workload has an impact call center system performance at peak time?, v) How satisfaction score is affected by other KPIs?. Hence, for a higher satisfaction rate, a connection established between the customer satisfaction surveys and the experienced process.

Agent errors such as "Agent rejection" during the call reduce customer satisfaction. When the staff does not have enough information to meet the customer's needs, they consult with a experienced person. Such cases are recorded with the "Waitlisted back" activity, which means that the call is put on hold and the call is returned after a while. In order to prevent these problems, call center staff can be

improved by providing information or supported by training activities.

The average duration of the processes is 3.8 minutes and 3.2 minutes for 5-point processes and other-point processes, respectively. One of the reasons for the high average duration is "Waitlisted" and "Waitlisted back" activities. In general, customers want their expectations and needs to be met quickly. However, according to the results, processes with a long average duration received 5 points from the customers. In the calls that have "Waitlisted" activity, customers may feel that the company tries more to solve the problems and conclude that they have cared more and valued. Therefore, it may have given 5 points in the satisfaction survey. In other words, the customer needs to solve the problem with the company. The satisfaction levels of the customers can be summarized as follows: When the customer's expectation from the service is negative, but the provided service is positive, the company obtains a high level of satisfaction from the experience of the customer. However, when the customer's expectations are positive, but the experienced service is negative, the company gets a low level of satisfaction from the customer.

Since average handle time is a tricky metric, we cannot say the call center system work well by only considering it. Average handle time requires to be in the adjusted range. If the handle time is too long, it may mean the agents struggle with customer's problem. Yet, when the average handle time is too short, it may mean that the agents cannot present any real support. Therefore satisfaction, as a direct way to measure real performance, is combined into the study. It is concluded that the impact of other two KPIs, average handle time and first call resolution (FCR) support satisfaction positively. Many former researches stated satisfaction is positively affected by FCR, handle time has a negative impact. However, this study showed handle time may also positively affect customer satisfaction in some situations.

The study has some limitations that should be improved in further researches. The attitude and communication of the agent towards the customer is a critical factor affecting satisfaction. However, there are unrecorded events such as the communication of staff in the data. This is a limitation of the study. The findings of the study can be validated by made some correction actions by considering the results of this study and then re-measuring the same performance indicator values. However, this can be regarded as a further improvement of the study because of the pandemic.

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